

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Understanding the requirements of the client and what the software needs to do.
2. Knowing how to test the software and identify problems or error.
3. Understanding the user needs, security and whether the software is suitable for its purpose.

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A Scenario A: A student writes a 50-line Python calculator.	Yes	No	The student is mainly writing code for a small program
B A team develops a payroll system used by 5,000 employees	Yes	Yes	The team needs to consider requirements, stakeholders testing and maintaining the system not just coding.

C An AI assistant generates a simple appointment application from one prompt.	Yes	No	AI can generate code but doesn't take in consideration of requirements, assumptions that still need to be checked.
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Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Receptionist/ clinic staff	Manage patients and appointments
Patients	Accurate patient and appointment records
Healthcare practitioners	View availability and appointments
Clinic management	View information and produce reports

Task 2 - Identify current problems

1. Difficulty location patient records
2. Duplicate appointment bookings
3. Limited visibility of practitioner availability
4. Inconsistent appointment status

Task 3 - Ask client questions

1. What patient information needs to be stored in the system?
2. What practitioner information needs to be stored?
3. How should duplicate bookings be prevented?
4. What appointment statuses are required?
5. What reports does management need?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Yes	Yes	Accept
Facial recognition login	No	No	Reject
AI diagnosis recommendations	No	No	Reject
Patient search	Yes	Yes	Accept
Online payment	No	No	Reject
Practitioner schedule view	Yes	Yes	Accept
Insurance processing	No	No	Reject
Treatment-plan generation	No	No	Reject

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

A software engineer must confirm the requirements with the client because AI may suggest features that the client did not ask for